CO544: Machine Learning and Data Mining
Lab 1: Python Data Science Toolbox

E/21/245
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Exercise 1: NumPy Advanced Operations

1. Insights and Observations

- ➔ Data Generation:- I generated a random dataset of 20 integers. This is useful for simulating real-world data distribution.
- ➔ Boolean Indexing:- Using a mask to filter values >= 50 demonstrated how to efficiently extract specific subsets of data without using slow loops.
- ➔ Broadcasting:- Adding a 1D array to a 2D array showed that NumPy can automatically align dimensions. This is an efficient way to perform element-wise operations on large datasets.
- ➔ Dot Product:- I calculated the dot product of two vectors, which is a fundamental operation in machine learning algorithms, such as calculating weighted sums in neural networks.

2. Strategies for handling Missing Data and Outliers

- ➔ Outlier Detection:- In this exercise, Boolean Indexing was used as a basic strategy to filter data. In a real-world scenario, I can use this same technique to identify and remove Outliers (extreme values) that fall outside a specific range
- ➔ Handling:- Once outliers are detected using a mask, I can either remove those rows or replace them with the mean/median value to ensure they do not negatively impact the model's accuracy.

Exercise 2: Matplotlib Subplots

- ➔ **Insights and Observations:-** In this exercise, I used `subplots` to compare Sine and Cosine waves side-by-side. This method is very useful for viewing different data trends at the same time. By sharing the x-axis and adding clear labels, the graphs became easy to read and understand. I also learned how to save the final plot as an image file for my report.
- ➔ **Strategies for Handling Outliers:-** I observed that plotting data is a great way to find outliers visually. If a data point does not follow the smooth curve of the wave, it can be identified as an error. To handle such outliers, I can use the filtering methods from the first exercise to remove them or replace them with a correct value.

Exercise 3: Pandas Cleaning & Preprocessing

- **Insights and Observations:-** In this exercise, I analyzed the Titanic dataset and discovered that it contained several issues, such as missing values and extreme outliers. I noticed that columns like 'Age' and 'Embarked' had incomplete data, which could lead to inaccurate analysis. Additionally, the 'Cabin' column had too many missing values to be useful, so I decided to drop it to simplify the dataset. This cleaning process is a vital step because high-quality data is necessary to build a reliable machine learning model.
- **Strategies for Handling Missing Data and Outliers:-** To handle missing data, I used Imputation. I replaced missing 'Age' values with the Median and missing 'Embarked' values with the Mode. I also used the Drop method to remove the 'Cabin' column entirely since it was mostly empty. For outliers in the 'Fare' column, I applied the IQR (Interquartile Range) method. By calculating the upper and lower bounds, I was able to identify and remove 116 records that fell outside the normal range. I also used a Boxplot to visually confirm that these outliers were removed, resulting in a cleaner and more balanced dataset.

Exercise 4: Pandas Essentials

- **Insights and Observations:-** In this exercise, I practiced the core functionality of Pandas by manipulating Series and DataFrames. By using `describe()`, I obtained a statistical summary of a random dataset (df2), noting that the values in column 'A' ranged from a minimum of -2.34 to a maximum of 2.85, with a mean of approximately 0.13. I also explored data selection techniques using `loc` and `iloc`. For instance, I successfully retrieved Alice's score of 85 using label-based indexing. Furthermore, I learned to sort data, such as sorting column 'A' in descending order, where the highest value was identified as 2.85. These tools are essential for exploring and organizing data efficiently.
- **Strategies for Handling Missing Data and Outliers:-** I explored two primary strategies for managing missing data (NaNs) in a DataFrame. First, I used `dropna()`, which removed any row containing a missing value, resulting in a DataFrame with only complete records (ex- keeping only index 2). Second, I used `fillna(0)`, which replaced all null values with 0, ensuring that the dataset remains the same size while allowing numerical operations to continue. Additionally, I practiced Excel I/O operations by reading a weather dataset. I observed that the weather data for late December showed maximum temperatures around 30.7°C, and I successfully exported an updated version of this file, which is a key step in maintaining data records.

Exercise 5: Loading Open Dataset from UCI Repository

- **Insights and Observations:-** In this exercise, I imported the Wine dataset from the UCI Repository, which contains chemical analysis results of wines grown in Italy. I observed that the dataset consists of 14 features, such as Alcohol, Magnesium, and Color Intensity. By inspecting the first 10 rows, I noticed that the 'Proline' levels vary significantly, with some values reaching as high as 1480. Loading data directly from a URL is a powerful way to work with standardized datasets in machine learning.
- **Interpretation of Group-by Results:-** I used the `groupby` function to calculate the mean values for each of the three wine classes. The results show clear differences between the classes. For example, Class 1 has the highest average Alcohol content at 13.74%, while Class 2 is the lowest at 12.28%. Additionally, Class 3 has a much higher Color Intensity (7.39) compared to Class 2 (3.08). These distinct statistical differences are important because they allow a machine learning model to accurately identify and classify the type of wine based on its chemical properties.

Exercise 6: scikit-learn Iris Dataset (Extended)

- **Insights from Visualization:-** The visual analysis using the **Pair Plot** and **Violin Plot** provided a clear explanation for the model's performance. The Pair Plot revealed that the **Setosa** species is completely isolated from the others, while **Versicolor** and **Virginica** show a slight overlap but remain mostly distinct. The **Petal Length** and **Petal Width** emerged as the most significant features for classification. The Violin Plot further confirmed these differences, showing that each species has a unique distribution for its physical dimensions.
- **Model Performance and Reflection:-** The **Logistic Regression** model achieved a perfect **Accuracy of 1.00**, with a Precision, Recall, and F1-score of 1.00 across all classes. While such a high score can sometimes indicate **Overfitting**, in this case, it is justified. Since the 100% accuracy was achieved on the **Test Set** (unseen data) and the species are naturally well-separated in the dataset, the model has successfully learned to generalize rather than just memorize. This exercise demonstrates that even a simple linear model can be highly effective when the data features are clearly defined and properly preprocessed.